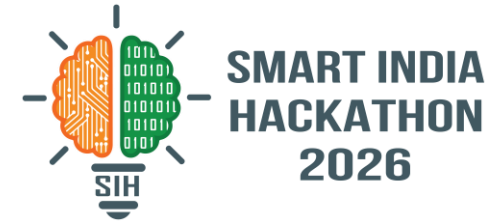


SMART INDIA HACKATHON 2026

SatQuery AI: Agentic Multimodal Remote Sensing Intelligence Platform



Problem Statement ID: SIH26167

Problem Statement Title: Agentic Multimodal AI for Remote Sensing Analysis (ISRO / SAC)

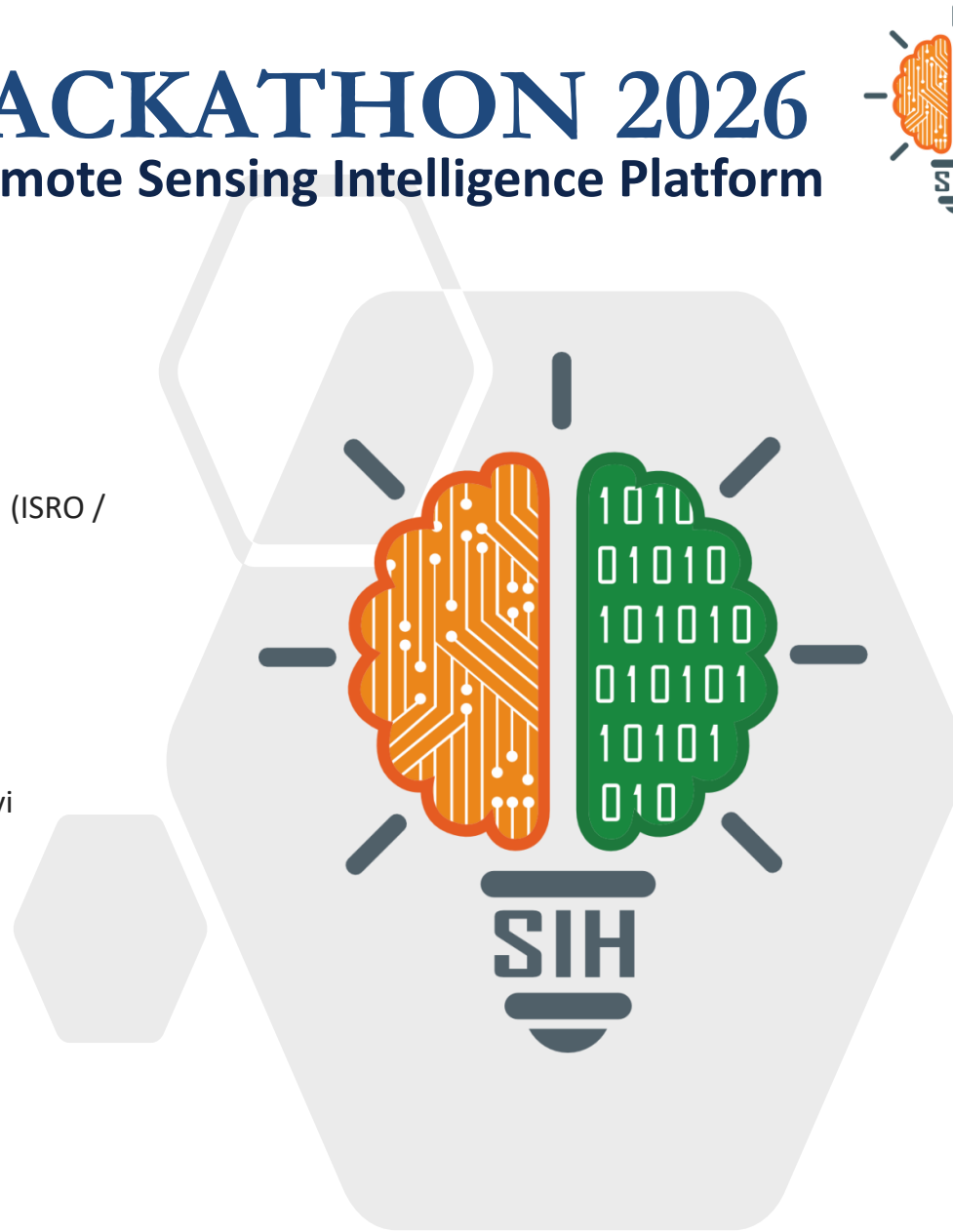
Theme: Space Technology / Disaster Management & Geospatial Intelligence

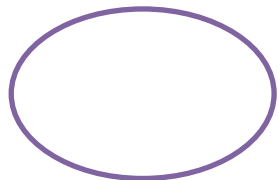
PS Category: Software

Team ID: [Registered Portal Team ID]

Team Name: SatQuery AI

Team Roster: Peter (Leader & Architect) | Pradipti (Research & Pitch) | Chhavi (AI/MLOps) | Vinayak (Frontend) | Achintya (Backend) | Misha (DB & GIS)





IDEA TITLE: SatQuery AI



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **SatQuery AI:** An agentic, multimodal Earth Observation platform designed for ISRO / Space Applications Centre (SAC) converting natural language into verifiable geospatial intelligence across Optical (Cartosat-2S, Sentinel-2) and SAR (RISAT-1A, Sentinel-1) imagery.
- **Interactive Web-GIS Application:** Features MapLibre GL dual-pane comparison swipe slider, dynamic GeoJSON vector polygon overlays, conversational AI chat drawer, and 1-click Intelligence Dossier PDF export.

Detailed explanation of the proposed solution

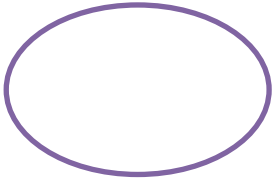
- **Deterministic Agentic Router (AgenticTaskRouter):** Dynamically classifies intent across 5 tasks (Single VQA, Captioning, SAM-2 Text Grounding, Bi-Temporal Change Detection, and Cross-Modal Optical-SAR Fusion) and validates spatial co-registration between image pairs.
- **Cross-Modal Optical-SAR Fusion:** Combines cloud-blind optical spectral bands with cloud-penetrating SAR microwave roughness (sigma0 in dB) for 24/7 all-weather flood inundation and border monitoring.
- **Zero Coordinate Hallucination:** Directly projects pixel masks via Affine Transformation Matrices into EPSG:4326 GeoJSON polygons with exact physical hectare calculations ($[lon, lat]^T = Affine * [x, y, 1]^T$).

How it addresses the problem

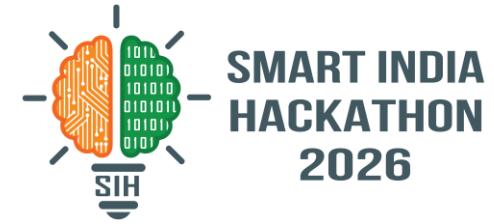
- **Solves Optical Cloud Blindspot:** Optical sensors are blind during monsoon cloud cover - precisely when flood and cyclone disaster intelligence is most urgent. SAR radar penetrates clouds to delineate standing water.
- **Democratizes Complex GIS:** Replaces cumbersome manual GIS desktop software with conversational English for disaster field officers.
- **100% Observable Telemetry:** Replaces opaque black-box LLMs with verifiable JSON execution traces detailing tools, bounds, and latency.

Innovation and uniqueness of the solution

- **Sovereign Dual-Use Architecture:** Tailored for ISRO Bhuvan / MOSDAC / NDMA disaster response + commercial agritech PMFBY crop insurance verification and NHAI infrastructure auditing.
- **Sub-2.5s Lightweight Inference:** 4-bit QLoRA fine-tuned Qwen2-VL-2B adapted for remote sensing, running in <6GB VRAM on consumer hardware.



TECHNICAL APPROACH

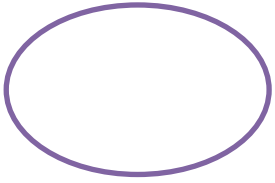


- **Technologies to be used (e.g. programming languages, frameworks, hardware)**

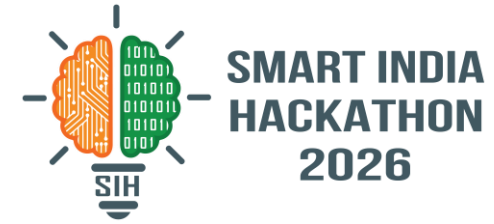
- **AI & MLOps Stack:** Python 3.11, PyTorch 2.6, Qwen2-VL-2B (4-bit QLoRA), Grounding DINO + SAM-2 (Segment Anything Model 2), ChangeFormer-V2 cross-attention head, ONNX Runtime, GPUManager (dynamic CUDA VRAM management & cache clearing).
- **Geospatial & Ingestion Engine:** Rasterio, GDAL, NumPy, SciPy (adaptive Lee speckle filter 5x5), Shapely, Affine transformation matrices.
- **Backend & Spatial Cache:** FastAPI REST Gateway, Pydantic v2 universal schemas, SQLite spatial cache (satquery_cache.db), ReportLab PDF engine.
- **Web-GIS Frontend:** React 18, TypeScript, MapLibre GL interactive canvas, Dual-Pane Split Swipe Slider, Tailwind CSS.

Methodology and process for implementation (Flow Charts/Images/ working prototype)

- **Step 1 (Multi-Modal Ingestion):** Parses GeoTIFFs (Cartosat, RISAT, Sentinel) and benchmark PNGs; verifies CRS, GSD, and band layouts; converts SAR amplitude to calibrated backscatter sigma0 (dB).
- **Step 2 (Agentic Task Orchestration):** AgenticTaskRouter parses natural language, determines task modality, verifies spatial co-registration between image footprints, and bounds parameters to strict ISRO guardrails (confidence, change threshold, filter kernel).
- **Step 3 (Dynamic Specialist Tool Dispatch):** Routes to RS-VQA-Engine, SpatialGrounding-Engine, BiTemporalChange-Engine, or OpticalSARFusion-Engine with microsecond telemetry timing.
- **Step 4 (Deterministic Affine Coordinate Projection):** Automatically projects binary prediction masks to Earth coordinates (EPSG:4326 GeoJSON polygons) and computes exact surface areas in hectares (ha).
- **Step 5 (Verifiable Output & Intelligence Dossier):** Emits observable JSON trace (AuditableExecutionTrace), renders interactive vector overlays, and compiles automated Intelligence Dossier PDF reports.



FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**

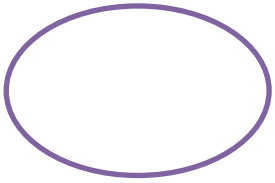
- **End-to-End Working Prototype Validated:** 36/36 automated integration tests passing (100% pass rate) on GitHub repo (Shourya3113/SatQuery-AI-SIH-26) across API, Orchestration, Geospatial Affine, and SQLite Registry modules.
- **Lean Hardware Footprint:** Modular specialist backends run within <6GB VRAM (operates smoothly on standard developer laptops and free Google Colab T4 GPUs, eliminating the 16GB VRAM barrier).
- **Sovereign & Air-Gapped Compatibility:** Entire platform operates self-contained without external third-party proprietary APIs (OpenAI/Anthropic); ready for deployment on NIC MeghRaj or ISRO Bhuvan Cloud servers.

Potential challenges and risks

- **Risk 1 (Spatial Misalignment):** Geometric distortion or CRS mismatches between multi-temporal passes or different sensor modalities (Optical vs SAR).
- **Risk 2 (SAR Speckle Noise):** Coherent radar interference creating false high-frequency edges and spurious change detections.
- **Risk 3 (LLM Coordinate Hallucination):** General generative models outputting fictitious geographic coordinates and arbitrary polygon boundaries.

Strategies for overcoming these challenges

- **Strategy 1 (Automated Preprocessing Guardrail):** Ingestion pipeline verifies CRS (EPSG:32633, EPSG:4326), bounding box overlap, and spatial resolution prior to tool dispatch.
- **Strategy 2 (Adaptive Lee Speckle Filtering):** Applied 5x5 window adaptive Lee filtering on calibrated SAR intensity to eliminate speckle noise while preserving sharp coastlines and urban edges.
- **Strategy 3 (Affine Matrix Coordinate Projection):** Hardcoded affine coordinate math directly computes polygon vertices from raster transformations, making coordinate hallucination mathematically impossible.



IMPACT AND BENEFITS

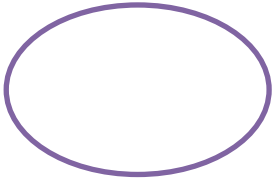


- **Potential impact on the target audience**

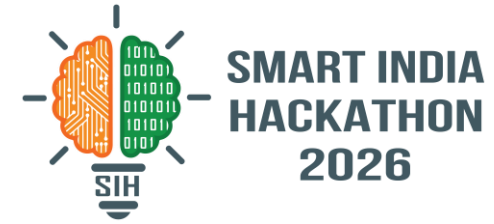
- **ISRO / SAC & National Disaster Management Authority (NDMA):** Instantaneous flood inundation and landslide mapping through dense clouds, cutting emergency damage assessment from days to <3 seconds.
- **Defense & Border Security Organizations:** 24/7 all-weather structural change detection and vehicle/runway monitoring along borders without manual GIS registration.
- **District Administration & Relief Field Officers:** Conversational natural-language interface enables field commanders to query satellite data without specialized GIS training.

Benefits of the solution (social, economic, environmental, etc.)

- **Social Benefit (Disaster Resilience):** Rapid, verified disaster mapping directly saves lives during monsoon floods, cyclones, and cloudburst events by directing first responders to isolated regions.
- **Economic Benefit (Dual-Use Commercial Market):**
 - Agritech & Crop Insurance: Automates crop damage assessment for Pradhan Mantri Fasal Bima Yojana (PMFBY), preventing fraudulent claims and expediting payouts.
 - Infrastructure Auditing: Dynamically monitors NHAI highway expansion, urban sprawl, and encroachments.
 - Cost Reduction: Sovereign open-source stack eliminates expensive commercial GIS desktop licenses (\$5,000+/seat).
- **Environmental Benefit:** Continuous monitoring of deforestation, reservoir depletion, and wetland conservation through automated NDVI/NDWI indexing and directional change analysis.



RESEARCH AND REFERENCES



- **Details / Links of the reference and research work**

- **Benchmark Datasets & Domain Adaptation:**

- BigEarthNet-MM: 590,326 Sentinel-1 SAR and Sentinel-2 Multispectral patch pairs used for cross-modal contrastive representation learning.
- VRSBench: High-resolution visual grounding, scene captioning, and remote sensing VQA benchmarks.
- RSVQA & CDVQA: High/Low resolution VQA and Change Detection Visual Question Answering benchmarks.
- ISRO Sensor Calibration: Cartosat-2S (0.65m GSD pan-sharpened optical) & RISAT-1A / EOS-04 (C-band SAR backscatter calibration).

- **Foundational Literature & Model Backbones:**

- Segment Anything Model 2 (SAM-2): Kirillov et al., Meta AI (2024) - Zero-shot promptable mask delineation.
- ChangeFormer-V2: Bandara & Patel (2022) - Bitemporal Transformer for remote sensing change detection.
- Qwen2-VL: Wang et al., Alibaba Cloud (2024) - Vision-Language model with dynamic resolution processing.

- **Project Repository & Verified Codebase:**

- GitHub Repository: <https://github.com/Shourya3113/SatQuery-AI-SIH-26>
- Test Suite: 36/36 automated integration tests passing across REST API, Orchestrator, Affine Engine, and SQLite Registry.